ĐẠI HỌC QUỐC GIA HÀ NỘI
TRƯỜNG ĐẠI HỌC CÔNG NGHỆ

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NGHIÊN CỨU VỀ ĐỘNG LỰC NỘI TẠI TRONG HỌC TĂNG CƯỜNG

KHOÁ LUẬN TỐT NGHIỆP

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RESEARCH ON INTRINSIC MOTIVATION FOR REINFORCEMENT LEARNING

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Declaration of Authorship

I hereby declare that this thesis is my own work and has not been submitted previously for a degree or diploma at this or any other higher education institution. To the best of my knowledge and belief, the thesis contains no material previously published or written by another person except where proper acknowledgment and citation are given.

Hanoi, ...

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Tóm tắt

Adaptive Correlation-Weighted Intrinsic Rewards for Reinforcement Learning nghiên cứu bài toán học tăng cường trong môi trường phần thưởng thưa, nơi tác tử rất khó khám phá do tín hiệu phản hồi từ môi trường xuất hiện không thường xuyên. Nhiều phương pháp hiện nay kết hợp phần thưởng ngoại sinh với phần thưởng nội tại dựa trên tò mò hoặc độ mới, nhưng thường phải dùng một hệ số cố định để cân bằng hai nguồn phần thưởng này. Hệ số cố định dễ nhạy với từng bài toán và không phản ánh được việc ở những trạng thái khác nhau thì nhu cầu khám phá cũng khác nhau.

Khóa luận trình bày ACWI, một cơ chế điều chỉnh phần thưởng nội tại theo từng trạng thái thông qua một Beta Network gọn nhẹ. Mạng này được huấn luyện bằng mục tiêu tương quan nhằm làm cho phần thưởng nội tại sau khi được scale có quan hệ chặt chẽ hơn với tổng phần thưởng ngoại sinh trong tương lai. Phương pháp được kết hợp với PPO và Intrinsic Curiosity Module để vừa tăng cường khả năng khám phá, vừa giữ được tính ổn định trong quá trình huấn luyện.

Kết quả thực nghiệm trên năm môi trường MiniGrid cho thấy ACWI cải thiện hiệu quả mẫu và giảm độ dao động so với PPO thuần túy cũng như PPO kết hợp ICM với hệ số cố định. Phương pháp đặc biệt hiệu quả trong các môi trường có phần thưởng thưa nhưng vẫn còn tín hiệu ngoại sinh mang tính định hướng. Trong những bài toán quá thưa, ACWI suy giảm một cách ổn định về gần hành vi của hệ số cố định thay vì gây mất ổn định cho quá trình học.

Từ khóa: học tăng cường, phần thưởng thưa, phần thưởng nội tại, khám phá dựa trên tò mò, điều chỉnh phần thưởng thích ứng

Abstract

Sparse-reward reinforcement learning remains difficult because agents receive limited feedback about whether their behavior is useful. A common solution is to augment the environment reward with intrinsic motivation signals such as curiosity or novelty, but most existing methods rely on a fixed coefficient to balance intrinsic and extrinsic rewards. This fixed weighting is highly sensitive to task characteristics and cannot distinguish between states where exploration is beneficial and states where it becomes distracting.

This thesis studies an adaptive alternative based on the paper *Adaptive Correlation-Weighted Intrinsic Rewards for Reinforcement Learning*. The proposed framework, ACWI, learns a state-dependent scaling factor through a lightweight Beta Network. The network is trained with a correlation-based objective that aligns scaled intrinsic rewards with discounted future extrinsic returns. In the implemented framework, ACWI is combined with Proximal Policy Optimization and the Intrinsic Curiosity Module, allowing the agent to adjust exploration pressure online while preserving training stability.

Experiments on five sparse-reward MiniGrid environments show that ACWI improves sample efficiency and reduces variance compared with PPO without intrinsic rewards and PPO with fixed intrinsic coefficients. The method is particularly effective in environments where sparse extrinsic rewards still provide occasional informative feedback. In extremely sparse settings, ACWI degrades gracefully to near-fixed scaling rather than becoming unstable. Overall, the thesis demonstrates that state-dependent intrinsic reward modulation is a practical and effective direction for improving exploration in reinforcement learning.

Keywords: reinforcement learning, sparse rewards, intrinsic motivation, curiosity-driven exploration, adaptive reward scaling

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List of Acronyms

Acronym	**Full Form**	**Description**
ACWI	Adaptive Correlation-Weighted Intrinsic	Proposed adaptive intrinsic reward scaling framework
RL	Reinforcement Learning	Learning paradigm based on sequential interaction with an environment
PPO	Proximal Policy Optimization	On-policy policy-gradient algorithm used as the training backbone
ICM	Intrinsic Curiosity Module	Intrinsic reward model based on forward prediction error
RND	Random Network Distillation	Curiosity-based method using predictor error against a random target network
RIDE	Rewarding Impact-Driven Exploration	Intrinsic reward method based on state-change impact
GAE	Generalized Advantage Estimation	Technique for reducing variance in policy-gradient advantage estimates
MDP	Markov Decision Process	Standard formal model for reinforcement learning environments

Chapter 1

Introduction

1.1 Background and Motivation

Reinforcement learning (RL) has become a central paradigm for training agents to solve sequential decision-making problems, with influential successes in Atari games, Go, and robotic control [1–3]. In many of these achievements, the learning process benefits from dense reward signals that provide frequent guidance about which behaviors are desirable. However, many practical environments are sparse-reward: the agent receives useful feedback only after a long sequence of actions, making exploration substantially harder [4].

To mitigate this problem, a large body of research augments the extrinsic reward supplied by the environment with an intrinsic reward that encourages exploration [5, 6]. Count-based methods reward the agent for visiting rarely explored states [7–9]. Curiosity-driven methods instead derive intrinsic rewards from prediction errors or representation discrepancies, as in the Intrinsic Curiosity Module (ICM) [10], Random Network Distillation (RND) [11], and Rewarding Impact-Driven Exploration (RIDE) [12]. These methods often improve exploration, but they still rely on a crucial design choice: how strongly intrinsic rewards should be weighted relative to extrinsic rewards.

In many existing approaches, the balance between the two reward sources is controlled by a fixed scalar coefficient. This choice is usually tuned manually for each environment, and the best value can vary significantly across tasks and training stages [13, 14]. A constant coefficient cannot distinguish between states where continued exploration is useful and states where exploration has become distracting. This limitation motivates a more adaptive mechanism that can modulate intrinsic rewards according to the current state and the long-term usefulness of exploration.

1.2 Research Problem

This thesis investigates the following problem: how can intrinsic rewards be scaled adaptively so that they promote useful exploration in sparse-reward reinforcement learning without destabilizing policy optimization?

More specifically, the thesis focuses on state-dependent reward modulation. Let the agent receive an extrinsic reward R_t^E from the environment and an intrinsic reward R_t^I from an exploration module. Standard methods optimize a shaped reward of the form

$$\bar{r}_t = R_t^E + \beta R_t^I, \tag{1.1}$$

where β is a fixed hyperparameter. The limitation of this formulation is that the same intrinsic scaling is applied uniformly to all states, even though the value of exploration may differ substantially across the state space.

The core research question of this thesis is therefore:

> Can we learn a state-dependent scaling factor for intrinsic rewards that is automatically adjusted online according to its relationship with future extrinsic return?

1.3 Research Objectives and Scope

The objective of the thesis is to adapt the paper *Adaptive Correlation-Weighted Intrinsic Rewards for Reinforcement Learning* into a thesis-style study and present a complete method for adaptive intrinsic reward scaling. The specific objectives are as follows:

- analyze the limitation of fixed intrinsic reward coefficients in sparse-reward reinforcement learning;
- review existing intrinsic motivation methods and adaptive reward weighting approaches;
- present ACWI, a framework that learns a state-dependent scaling factor through a lightweight Beta Network;
- describe how ACWI is integrated with PPO and ICM for practical training;

- evaluate the method on sparse-reward MiniGrid environments and discuss its strengths and limitations.

The scope of the thesis is limited to episodic RL tasks with sparse rewards, using PPO as the base policy optimization algorithm and ICM as the intrinsic reward generator. The experiments focus on benchmark MiniGrid environments rather than real-world robotics or continuous-control tasks. Likewise, this thesis emphasizes empirical performance and methodological clarity instead of a full theoretical proof for convergence or optimality.

1.4 Research Methodology

The work follows a research-oriented methodology. First, the thesis surveys foundational RL literature and exploration methods to identify the gap addressed by the proposed approach. Second, it formulates ACWI as an adaptive scaling mechanism that predicts a positive coefficient $\beta(s_t)$ from the current state. Third, the method is integrated into the PPO training loop with ICM-derived intrinsic rewards. Finally, the approach is validated experimentally on multiple sparse-reward tasks in MiniGrid by comparing it against PPO without intrinsic rewards and PPO with ICM under several fixed coefficients.

The evaluation criteria emphasize sample efficiency, learning stability, robustness across environments, and the qualitative behavior of the learned adaptive scaling policy. This combination of theoretical grounding, algorithmic design, and empirical validation is consistent with the structure of a machine learning graduation thesis.

1.5 Thesis Structure

The remainder of this thesis is organized as follows. Chapter 2 reviews reinforcement learning with intrinsic motivation, sparse-reward exploration methods, and prior attempts to adapt the weighting of intrinsic rewards. Chapter 3 presents the proposed methodology, including the background formulation, the ICM module, the Adaptive Correlation-Weighted Intrinsic (ACWI) framework, and the training algorithm. Chapter 4 describes the experimental setup and analyzes the results on MiniGrid benchmark environments. Finally, the conclusion summarizes the main findings, discusses limitations, and outlines future research directions.

Chapter 2

Literature Review

2.1 Reinforcement Learning in Sparse-Reward Environments

An episodic reinforcement learning problem is commonly modeled as a Markov Decision Process (MDP) defined by the tuple $(\mathcal{S}, \mathcal{A}, P, R, \gamma, \rho_0)$, where $\mathcal{S}$ is the state space, $\mathcal{A}$ is the action space, P denotes the transition dynamics, R is the reward function, $\gamma \in [0, 1)$ is the discount factor, and ρ_0 is the initial state distribution [3]. The goal is to learn a policy $\pi_\theta(a \mid s)$ that maximizes the expected discounted return:

$$J(\theta) = \mathbb{E}_{\tau \sim (\pi_\theta, P)} \left[\sum_{t=0}^{T-1} \gamma^t R_t^E \right]. \tag{2.1}$$

Although this formulation is standard, learning becomes difficult when rewards are sparse. In such cases, the agent may take many actions before receiving any meaningful signal indicating whether the trajectory is promising. Classical exploration strategies such as ϵ-greedy can provide randomization, but they do not explicitly guide the agent toward informative experiences when the state space is large or partially observable [4].

2.2 Intrinsic Motivation for Exploration

Intrinsic motivation addresses sparse rewards by supplementing the environment reward with an additional exploration bonus [5, 6]. In general, the agent receives both an extrinsic reward R_t^E and an intrinsic reward R_t^I, and optimization is performed on a combined objective:

$$\bar{r}_t = R_t^E + \beta R_t^I, \tag{2.2}$$

where $\beta \geq 0$ controls the importance of intrinsic motivation.

2.2.1 Count-Based Exploration

Count-based methods encourage exploration by assigning larger rewards to states that have been visited less frequently. In tabular settings, the intrinsic reward is often inversely proportional to the square root of the visit count. This idea has been extended to high-dimensional observations through pseudo-counts and density models [7–9]. These approaches are conceptually simple and often effective, but the quality of the reward signal depends on the ability to estimate novelty reliably in complex observation spaces.

2.2.2 Curiosity-Driven Exploration

Curiosity-driven methods define intrinsic rewards through prediction error. The Intrinsic Curiosity Module (ICM) learns a compact state representation and predicts the next latent feature from the current feature and action [10]. The prediction error of this forward model becomes an intrinsic bonus. Random Network Distillation (RND) uses the discrepancy between a fixed random target network and a learned predictor as a novelty measure [11]. RIDE rewards changes in a learned representation that are causally induced by the agent's actions [12]. A common advantage of these methods is that intrinsic rewards tend to decrease naturally as states become familiar, reducing repeated exploration of the same region.

2.3 Adaptive Weighting of Intrinsic Rewards

Despite the success of intrinsic motivation, many methods still depend on a manually chosen coefficient β [10, 11]. This fixed coefficient creates an important limitation. If β is too small, the intrinsic reward becomes negligible and exploration remains weak. If β is too large, curiosity may dominate the training signal and distract the agent from the task objective. Since the optimal trade-off differs across environments and over the course of training, fixed weighting lacks flexibility.

Several studies have attempted to overcome this issue. Extrinsic-Intrinsic Policy Optimization (EIPO) treats the trade-off as a constrained optimization problem and adapts the influence of intrinsic rewards at a global level [13]. Automatic Intrinsic Reward Shaping (AIRS) frames the selection among different intrinsic reward functions as a bandit-style decision process [14]. Population-based approaches such as Never Give Up and Agent57 coordinate multiple policies with different exploration strengths and discount factors [15, 16]. These methods improve adaptability, but their modulation is usually global, policy-level, or stage-level rather than state-dependent.

2.4 Research Gap

The literature suggests that intrinsic motivation is effective when the exploration bonus is aligned with future task success. However, existing methods often lack a direct mechanism for deciding *where* intrinsic rewards should be emphasized. Two states can have similar novelty scores while contributing very differently to downstream extrinsic return. A uniform or globally adapted coefficient cannot express this difference.

The central gap addressed in this thesis is therefore the absence of a lightweight, state-dependent scaling mechanism that modulates intrinsic rewards according to how well they predict future extrinsic outcomes. The ACWI framework fills this gap by learning a scaling function $\beta(s_t)$ and optimizing it using a correlation-based objective. This design allows the agent to strengthen exploration when it is likely to be beneficial and reduce it when exploitation should dominate.

2.5 Chapter Summary

This chapter reviewed the foundations required for the proposed work. Reinforcement learning in sparse-reward environments is challenging because standard exploration methods provide weak guidance. Intrinsic motivation methods such as count-based bonuses, ICM, RND, and RIDE improve exploration but usually depend on a fixed weighting coefficient. Prior adaptive methods alleviate this problem only partially because they operate at a coarse granularity. These observations motivate the methodology presented in the next chapter, where intrinsic reward scaling is learned directly as a state-dependent function.

Chapter 3

Methodology

3.1 Problem Formulation

This thesis considers episodic reinforcement learning with both extrinsic and intrinsic rewards. At time step t, the agent observes state s_t, samples action $a_t \sim \pi_\theta(\cdot \mid s_t)$, receives extrinsic reward R_t^E, and transitions to the next state s_{t+1}. Let R_t^I denote the intrinsic reward produced by an exploration module. Instead of using a constant coefficient, the proposed method introduces a state-dependent scaling function $\beta_\psi(s_t)$ so that the shaped reward becomes

$$\bar{r}_t = R_t^E + \alpha\beta_\psi(s_t)R_t^I, \tag{3.1}$$

where $\alpha > 0$ is a global intrinsic-strength hyperparameter and $\beta_\psi(s_t)$ is learned online.

The intuition is straightforward. If exploration from state s_t tends to lead to higher future extrinsic return, the method should increase the influence of intrinsic reward in that state. If exploration is unhelpful or noisy, the influence should be reduced. ACWI operationalizes this idea through a lightweight Beta Network trained with a correlation-based objective.

3.2 Base Reinforcement Learning Algorithm

The policy optimization backbone used in this thesis is Proximal Policy Optimization (PPO) [17]. PPO is widely adopted because it offers stable on-policy optimization with relatively simple implementation. It updates the policy by maximizing a clipped surrogate objective:

$$L^{\mathrm{PPO}}(\theta) = \mathbb{E}_t\left[\min\left(\rho_t(\theta)\hat{A}_t, \mathrm{clip}(\rho_t(\theta), 1-\epsilon, 1+\epsilon)\hat{A}_t\right)\right], \tag{3.2}$$

where $\rho_t(\theta) = \frac{\pi_\theta(a_t|s_t)}{\pi_{\theta_{\mathrm{old}}}(a_t|s_t)}$ is the importance ratio and $\hat{A}_t$ is the advantage estimate. The advantages are computed with Generalized Advantage Estimation (GAE) [18] using the shaped reward in Equation 3.1.

3.3 Intrinsic Curiosity Module

To generate intrinsic rewards, the thesis adopts the Intrinsic Curiosity Module (ICM) [10]. ICM first encodes each state into a latent representation $\phi(s_t) \in \mathbb{R}^d$. It then uses two auxiliary models:

- a forward dynamics model f_η^F that predicts the next latent feature from the current feature and action;
- an inverse dynamics model f_η^I that predicts the action from two consecutive latent states.

The ICM loss is a weighted combination of the forward and inverse losses:

$$L_{\mathrm{ICM}}(\eta) = \alpha_F \mathbb{E}\left[\frac{1}{2}\left\|f_\eta^F(\phi_t, a_t) - \phi_{t+1}\right\|_2^2\right] + \alpha_I \mathbb{E}\left[-\log p_\eta(a_t \mid \phi_t, \phi_{t+1})\right]. \tag{3.3}$$

The raw intrinsic reward is defined as the forward prediction error:

$$I_t = \frac{1}{2}\left\|f_\eta^F(\phi_t, a_t) - \phi_{t+1}\right\|_2^2. \tag{3.4}$$

To improve stability, the reward is standardized within each minibatch and then rectified:

$$I_t^+ = \max\left(0, \frac{I_t - \mu_I}{\sigma_I + \varepsilon}\right), \tag{3.5}$$

where μ_I and σ_I are the minibatch mean and standard deviation. This preprocessing suppresses transitions whose prediction errors are below the batch average and retains relatively surprising transitions.

3.4 Adaptive Correlation-Weighted Intrinsic Scaling

3.4.1 Beta Network

The core contribution of the thesis is a Beta Network $\beta_\psi : \mathcal{S} \to \mathbb{R}_+$ that predicts a positive intrinsic reward multiplier from the current state. The network is intentionally lightweight to keep the computational overhead low. In the experiments adapted from the paper, the network uses an encoder depth of 2, an embedding size of 256, and clips its output to the interval $[0.1, 2.0]$.

The scaled intrinsic reward is

$$\tilde{I}_t = \beta_\psi(s_t) I_t^+, \tag{3.6}$$

and the final reward used by PPO is

$$\bar{r}_t = R_t^E + \alpha \tilde{I}_t. \tag{3.7}$$

3.4.2 Correlation-Based Objective

Training β_ψ with meta-gradients would be computationally expensive. Instead, ACWI introduces a first-order objective that directly aligns the scaled intrinsic reward with discounted future extrinsic return. Let

$$G_t^E = \sum_{k=t}^{T} \gamma^{k-t} R_k^E \tag{3.8}$$

denote the discounted extrinsic return from time step t onward.

Within a minibatch, both the scaled intrinsic reward and the extrinsic return are standardized:

$$\hat{I}_t = \frac{\tilde{I}_t - \mu_{\tilde{I}}}{\sigma_{\tilde{I}}}, \qquad \hat{G}_t = \frac{G_t^E - \mu_G}{\sigma_G}. \tag{3.9}$$

The correlation loss is then defined as

$$L_{\text{corr}}(\psi) = -\mathbb{E}[\hat{I}_t \hat{G}_t]. \tag{3.10}$$

Minimizing this loss encourages $\beta_\psi(s_t)$ to be larger in states where intrinsic exploration is positively associated with future extrinsic success.

To avoid unstable or degenerate scaling, ACWI also uses a regularization term in log-space:

$$L_{\text{reg}}(\psi) = \mathbb{E}\left[\left(\log \beta_\psi(s_t) - \log \beta_0\right)^2\right], \tag{3.11}$$

where $\beta_0 = 1.0$ is the reference value. The complete objective is

$$L_\beta(\psi) = L_{\text{corr}}(\psi) + \lambda_{\text{reg}} L_{\text{reg}}(\psi). \tag{3.12}$$

3.5 Training Procedure

The complete training loop integrates PPO, ICM, and ACWI as follows:

1. collect on-policy trajectories using the current policy;
2. update the ICM module on the collected batch to compute intrinsic rewards;
3. standardize and rectify the intrinsic reward signal;
4. compute discounted extrinsic returns for each time step;
5. update the Beta Network by minimizing the correlation-based objective;
6. construct the shaped reward with the learned state-dependent scaling;
7. compute GAE advantages from the shaped reward;
8. update the policy and value networks with PPO for several epochs.

This procedure preserves compatibility with conventional PPO training while adding only a small auxiliary network and a simple optimization step before each PPO update cycle.

3.6 Discussion

The methodology offers three important advantages. First, it avoids manual selection of a single intrinsic reward coefficient for all states. Second, it remains computationally efficient because it does not require unrolled policy optimization or high-order differentiation. Third, it is modular: the scaling mechanism is agnostic to the base RL algorithm and can be paired with other intrinsic reward generators besides ICM.

At the same time, the method assumes that at least some correlation exists between local exploration and future extrinsic return. When the environment is so sparse that extrinsic signals remain near zero for long periods, the Beta Network receives little informative supervision. In such cases, the method is expected to behave similarly to a fixed scaling strategy rather than producing strong adaptation. This limitation becomes important in the empirical analysis of Chapter 4.

Chapter 4

Experimental Setup and Results

4.1 Benchmark Environments

The empirical evaluation is conducted on five sparse-reward MiniGrid environments [19]. These environments are chosen because they represent different forms of exploration difficulty while maintaining a common gridworld interface.

Bảng 4.1: MiniGrid environments used in the experiments

Environment	**Purpose in the evaluation**
DoorKey-8x8	Tests whether the agent can discover a key, unlock a door, and shift from exploration to goal-directed behavior.
Empty-16x16	Represents an extreme sparse-reward setting with almost no intermediate feedback before the goal is reached.
RedBlueDoors-8x8	Requires opening two doors in the correct order, testing whether the method can suppress irrelevant exploration.
UnlockPickup	Extends DoorKey with an additional object manipulation stage, creating a hierarchical sparse-reward task.
KeyCorridorS3R3	A longer-horizon task with multiple rooms, keys, and locked doors, used to assess sustained exploration ability.

These tasks share several useful properties: sparse terminal rewards, egocentric partial observations, step penalties that discourage random motion, and stochastic initialization over multiple random seeds. Together, they form a suitable test bed for evaluating whether adaptive intrinsic reward scaling improves both efficiency and robustness.

4.2 Baselines and Evaluation Criteria

The proposed ACWI method is compared against two categories of baselines:

- PPO trained only with extrinsic rewards;
- PPO combined with ICM using fixed intrinsic coefficients $\beta \in \{0.1, 0.2, 0.5, 1, 2\}$.

This comparison is important because it isolates the contribution of adaptive scaling from the contribution of intrinsic motivation itself. PPO provides a lower bound on exploration ability, while fixed-coefficient PPO+ICM shows how sensitive performance is to manual tuning.

The evaluation focuses on four aspects:

- episode return over training time;
- sample efficiency in the early and middle stages of learning;
- variance across random seeds;
- qualitative behavior of the learned scaling coefficient $\beta(s)$.

4.3 Experimental Configuration

All methods share the same PPO backbone and ICM architecture so that the comparison is fair. The adapted paper fixes the global intrinsic coefficient at $\alpha = 0.001$ and clips the learned Beta Network output to $[0.1, 2.0]$. The Beta Network uses a hidden representation size of 256 with encoder depth 2. The regularization weight for the correlation objective is $\lambda_{\text{reg}} = 10^{-3}$, the reference value is $\beta_0 = 1.0$, gradient clipping is set to 1.0, and weight decay is 10^{-6}.

Bảng 4.2: Key hyperparameters reported for ACWI

Hyperparameter	**Value**
Intrinsic strength α	0.001
Beta output bounds	$[0.1, 2.0]$
Encoder size	256
Encoder depth	2
Regularization weight λ_{reg}	10^{-3}
Reference value β_0	1.0
Gradient clipping	1.0
Weight decay	10^{-6}

The results are reported over five random seeds. This setup is sufficient to assess whether adaptive scaling consistently improves learning behavior rather than benefiting from a single favorable initialization.

4.4 Results

The experiments show several consistent trends. First, ACWI performs strongly across environments with sparse but still informative extrinsic rewards. In DoorKey-8x8, RedBlueDoors-8x8, UnlockPickup, and KeyCorridorS3R3, the adaptive approach achieves faster learning and lower variance than most fixed-coefficient baselines. This result indicates that ACWI improves sample efficiency and makes exploration behavior more stable across runs.

Second, the comparison against fixed-coefficient PPO+ICM highlights the difficulty of manual tuning. A coefficient that works well in one environment can be too small or too large in another. Small values underemphasize exploration and slow down learning, while large values can inject excessive noise and interfere with exploitation. ACWI mitigates both problems by learning to amplify intrinsic rewards only in states where they correlate with future extrinsic success.

Third, the method exhibits a natural transition from exploration to exploitation. As training progresses and the policy begins to discover task-relevant behaviors, the effective intrinsic contribution tends to decrease. This suggests that ACWI does not merely add curiosity-driven noise, but instead learns to reduce exploration pressure once the task structure becomes clearer.

4.5 Analysis of the Learned Scaling Behavior

The paper reports that the learned distribution of $\beta(s)$ evolves differently across environments. In DoorKey-8x8 and RedBlueDoors-8x8, the distribution is initially concentrated near its reference value, then gradually broadens and becomes more structured. This pattern suggests that the Beta Network learns to partition the state space into regions where different exploration intensities are appropriate. Toward the end of training, the distribution shifts toward lower values, reflecting the decreasing marginal value of intrinsic rewards once the policy has matured.

In contrast, Empty-16x16 behaves differently. Because extrinsic rewards remain almost zero for most trajectories, the correlation objective receives very weak supervision. As a result, the Beta Network stays close to its initialization and exhibits limited adaptation. Importantly, this failure mode is graceful rather than unstable: the method effectively defaults to a nearly fixed intrinsic coefficient instead of diverging. This observation confirms both a strength and a limitation of ACWI. The method is robust when correlation signals are weak, but meaningful adaptation requires at least occasional extrinsic feedback.

4.6 Discussion

Overall, the empirical evidence supports the main claim of the thesis: state-dependent intrinsic reward scaling can improve sparse-reward reinforcement learning when the environment provides enough structure for the agent to relate exploration to eventual success. ACWI is especially beneficial in environments where exploration needs change across states and over time. The method is less effective in extremely sparse tasks where the extrinsic return is too weak to supervise adaptation. Therefore, ACWI should be viewed as a practical middle ground between fixed intrinsic scaling and more expensive meta-learning approaches.

Conclusion and Future Work

This thesis presented an English thesis-style adaptation of the paper *Adaptive Correlation-Weighted Intrinsic Rewards for Reinforcement Learning*. The work addressed the limitation of fixed intrinsic reward coefficients in sparse-reward reinforcement learning and introduced ACWI, a lightweight framework that predicts a state-dependent scaling factor for intrinsic rewards. By aligning the scaled intrinsic signal with discounted future extrinsic return, ACWI provides a practical way to balance exploration and exploitation online without relying on expensive meta-gradient procedures.

The experimental study on MiniGrid environments shows that ACWI improves learning stability and sample efficiency in several sparse-reward tasks, especially when extrinsic signals are infrequent but still informative. Compared with manually tuned fixed coefficients, the adaptive mechanism is more robust across environments and random seeds. At the same time, the results also reveal an important limitation: when extrinsic rewards are almost always absent, the correlation objective becomes weak and the learned scaling factor behaves similarly to a fixed coefficient.

Several directions can extend this work in the future. First, ACWI can be integrated with intrinsic reward modules beyond ICM, such as RND or RIDE. Second, the method can be tested in broader settings including continuous control, multi-task reinforcement learning, and embodied navigation. Third, a more formal theoretical analysis of the correlation objective could clarify when the adaptive scaling mechanism is guaranteed to be beneficial. These directions would further strengthen the role of adaptive intrinsic reward weighting in reinforcement learning research.

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